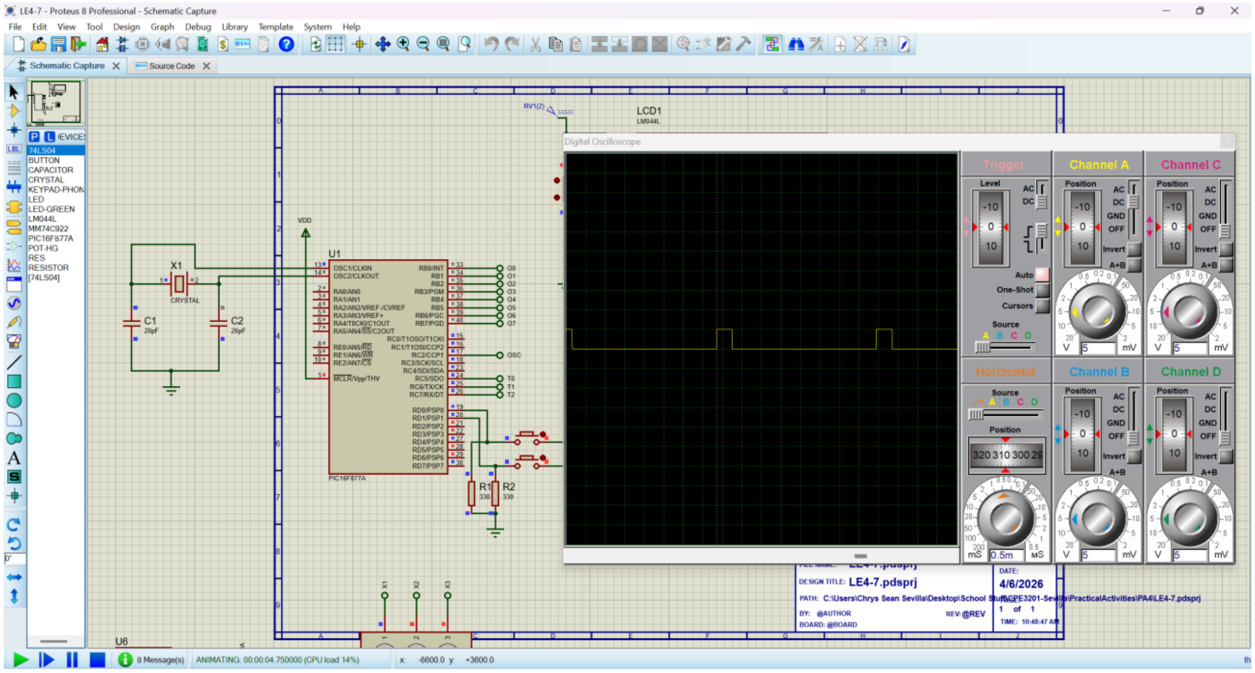
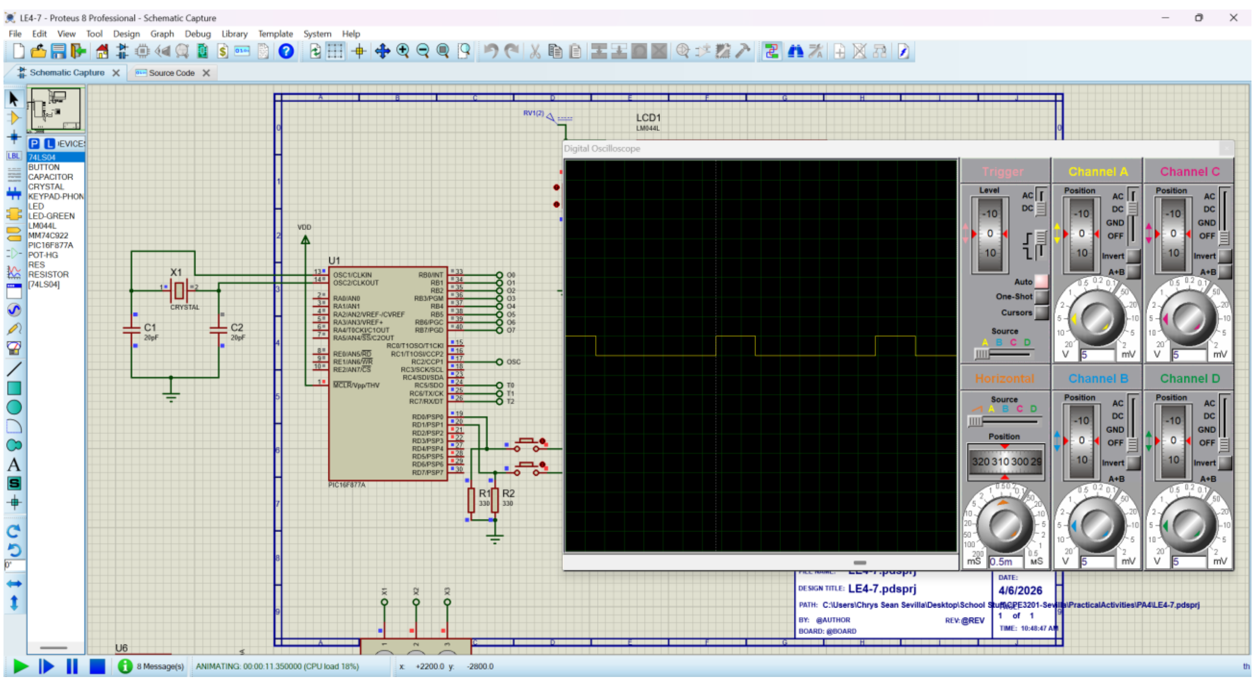
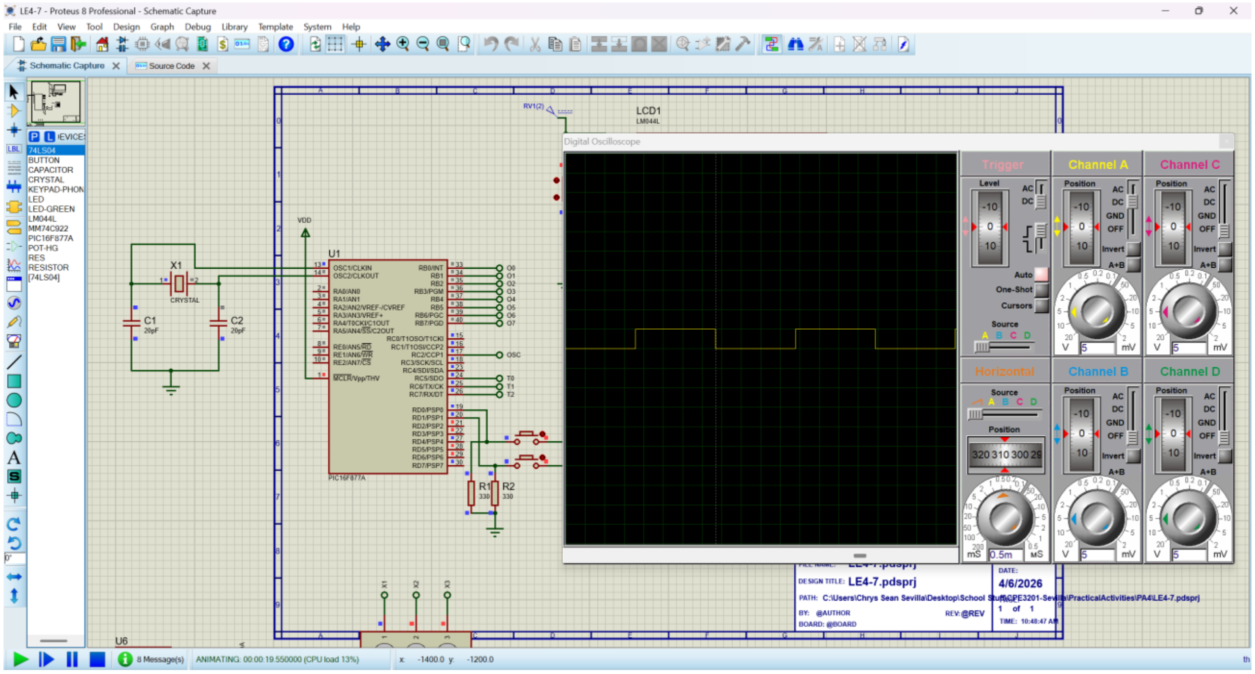
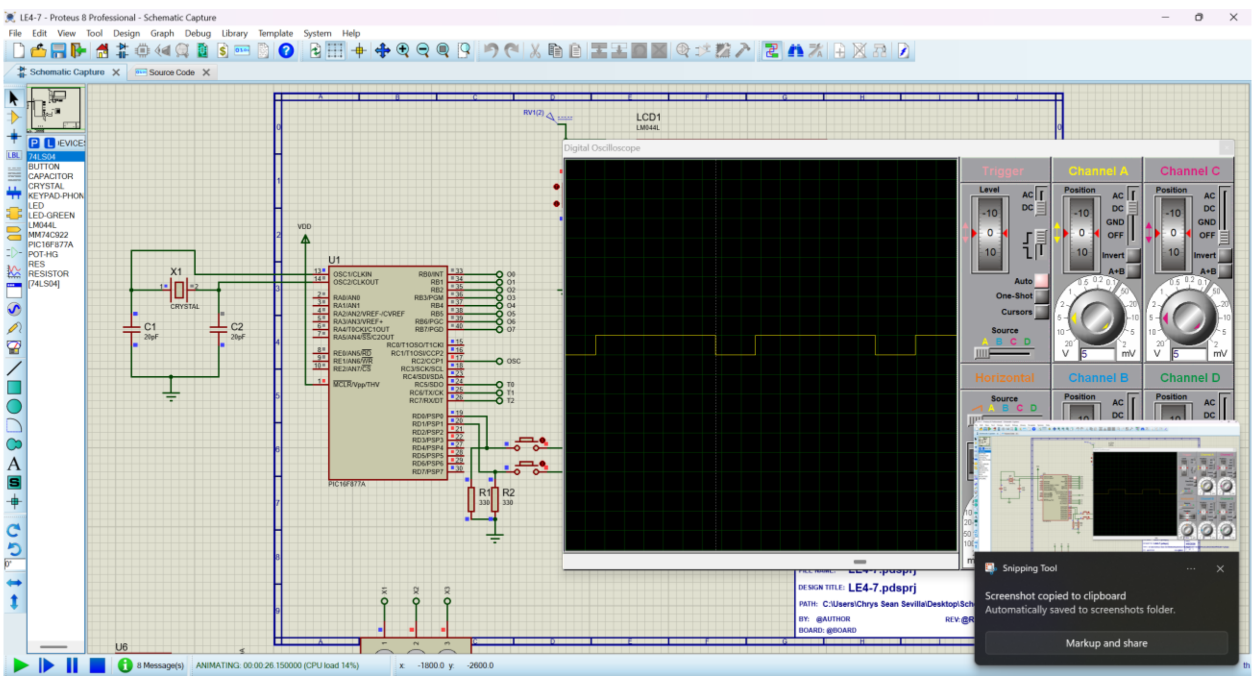
Part I






LE4-7 - Proteus 8 Professional - Schematic Capture
File Edit View Tool Design Graph Debug Library Template System Help
Schematic Capture
Source Code
Digital Oscilloscope
Trigger
Channel A
Channel C
Horizontal
Channel B
Channel D
LCD1
LM044L
U1
PIC16F877A
X1
CRYSTAL
C1
C2
R1
R2
DESIGN TITLE: LE4-7.pdsprj
DATE: 4/6/2026
ANIMATING: 00:00:19.550000 (CPU load 13%)


LE4-7 - Proteus 8 Professional - Schematic Capture
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Horizontal
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LCD1
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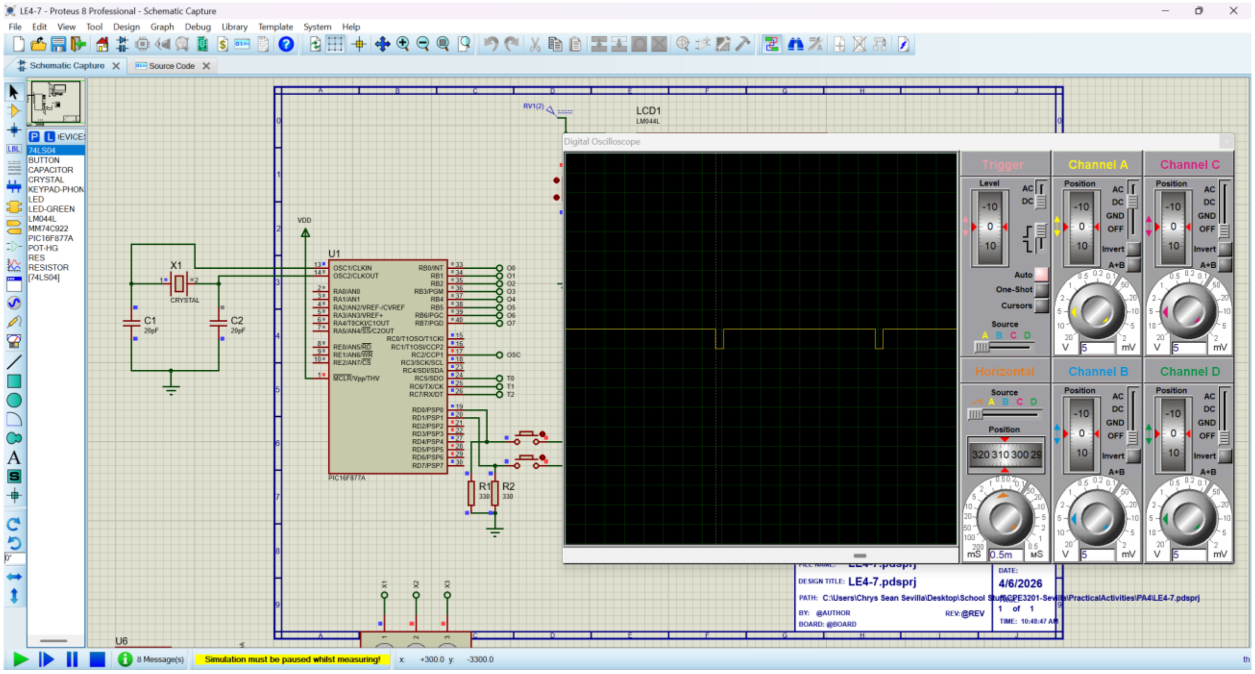
PIC16F877A
X1
CRYSTAL
C1
C2
R1
R2
DESIGN TITLE: LE4-7.pdsprj
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Markup and share
ANIMATING: 00:00:26.150000 (CPU load 14%)


LE4-7 - Proteus 8 Professional - Schematic Capture
File Edit View Tool Design Graph Debug Library Template System Help
Schematic Capture
Source Code
Digital Oscilloscope
Trigger
Channel A
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Channel C
Channel D
Horizontal
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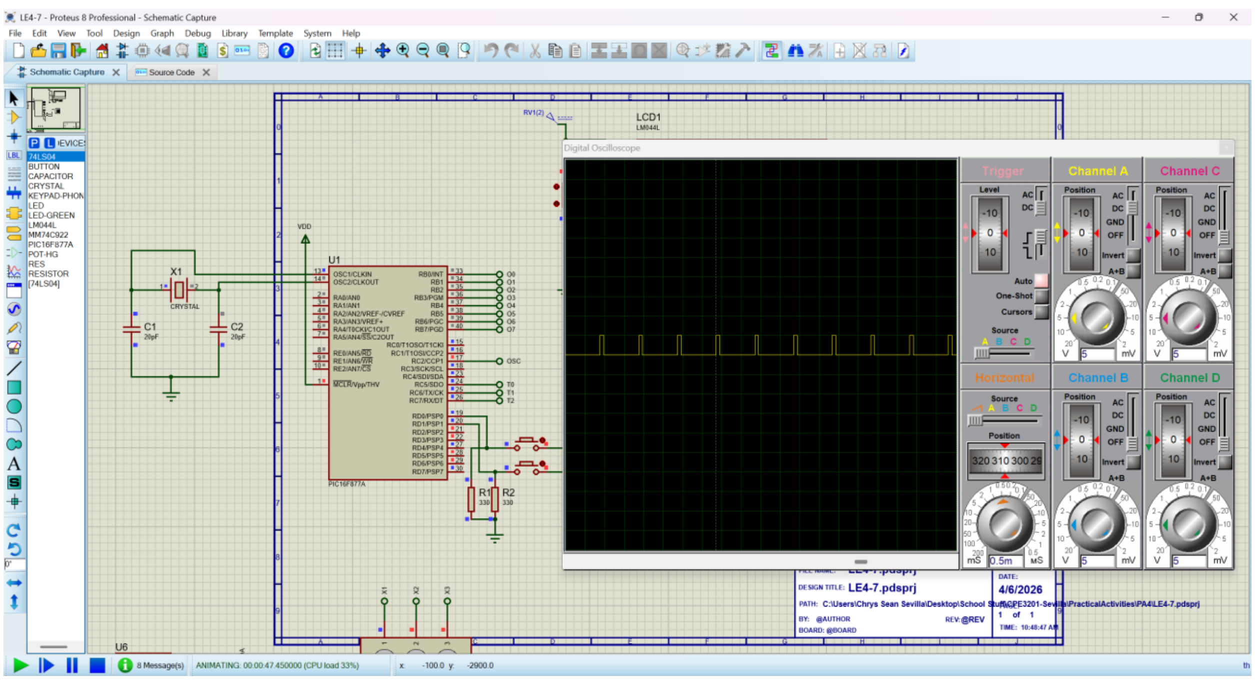
U1
PIC16F877A
X1
CRYSTAL
C1
C2
R1
R2
DESIGN TITLE: LE4-7.pdsprj
DATE: 4/6/2026
8 Message(s)
Simulation must be paused whilst measuring!


LE4-7 - Proteus 8 Professional - Schematic Capture
File Edit View Tool Design Graph Debug Library Template System Help
Schematic Capture
Source Code
Digital Oscilloscope
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Channel A
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Channel C
Channel D
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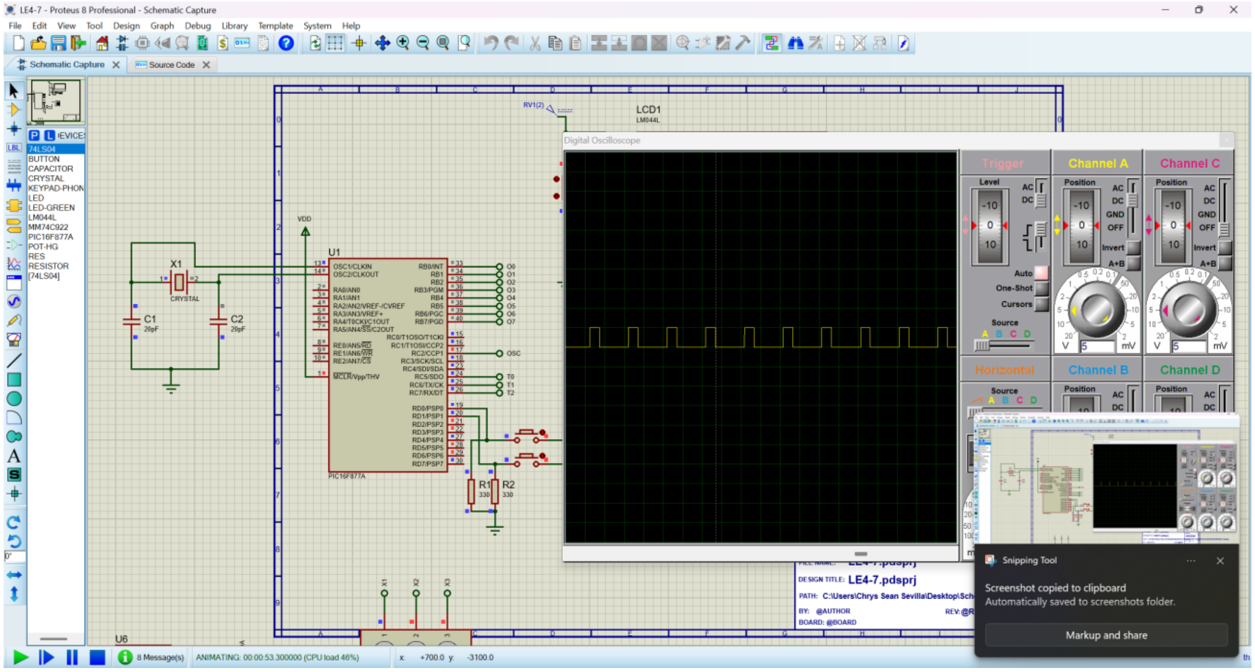
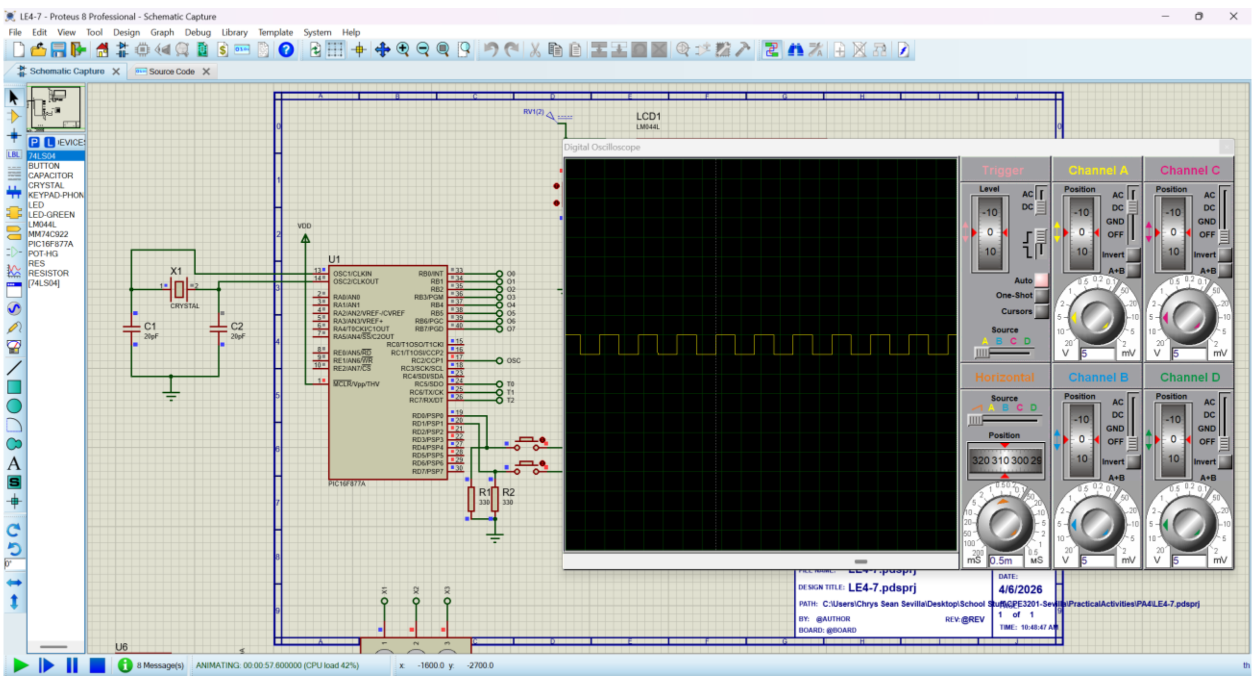
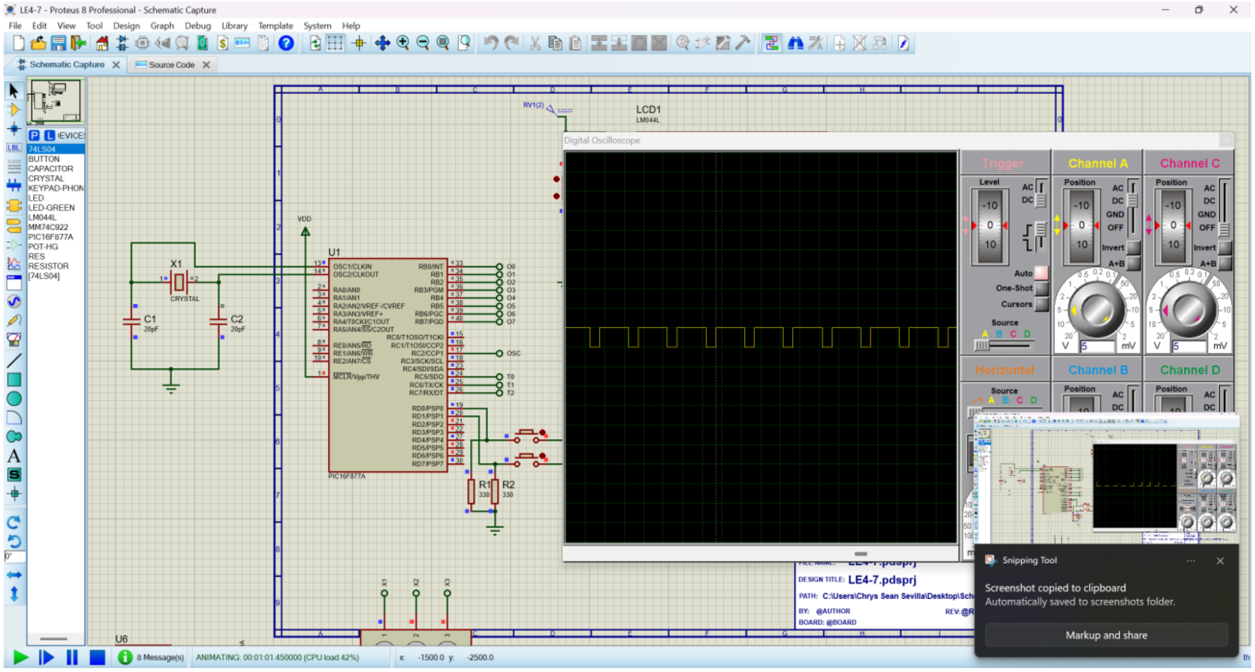
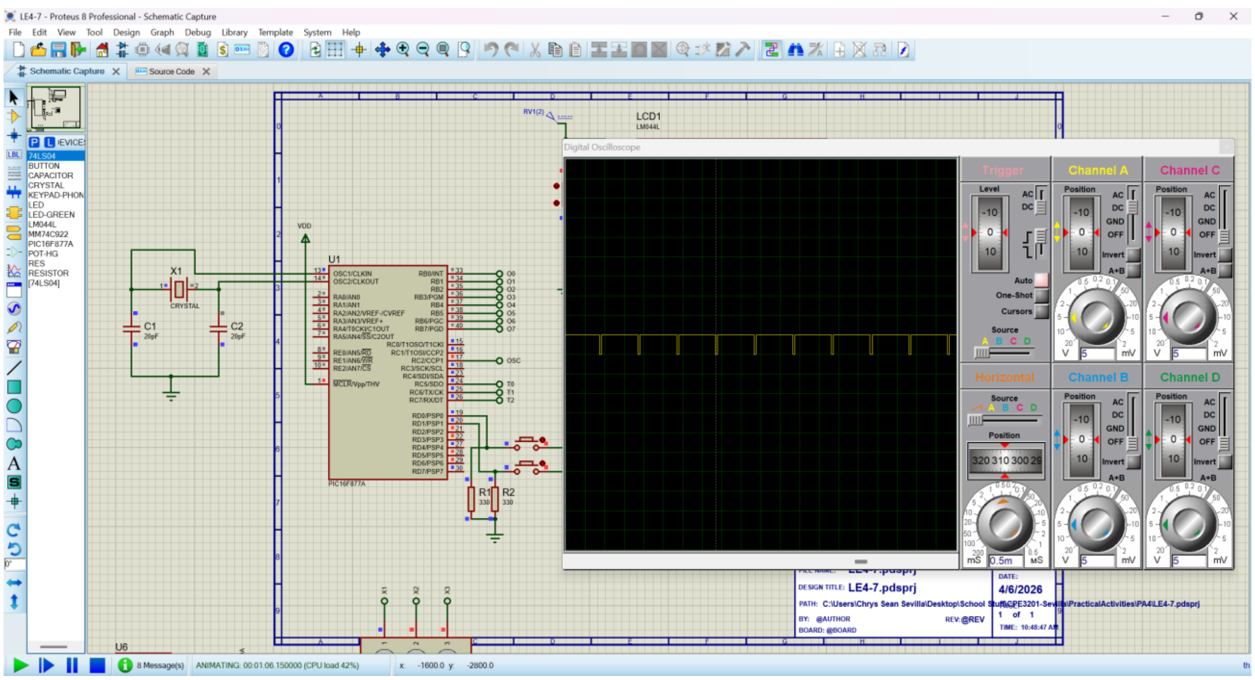
PIC16F877A
X1
CRYSTAL
C1
C2
R1
R2
DESIGN TITLE: LE4-7.pdsprj
DATE: 4/6/2026
8 Message(s)
ANIMATING: 00:00:47.450000 (CPU load 33%)


LE4-7 - Proteus 8 Professional - Schematic Capture
File Edit View Tool Design Graph Debug Library Template System Help
Schematic Capture
Source Code
Digital Oscilloscope
Trigger
Channel A
Channel C
Horizontal
Channel B
Channel D
LCD1
U1
PIC16F877A
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Markup and share
DESIGN TITLE: LE4-7.pdsprj
8 Message(s)
ANIMATING: 00:00:53.300000 (CPU load 46%)


LE4-7 - Proteus 8 Professional - Schematic Capture
File Edit View Tool Design Graph Debug Library Template System Help
Schematic Capture
Source Code
Digital Oscilloscope
Trigger
Channel A
Channel C
Horizontal
Channel B
Channel D
LCD1
U1
PIC16F877A
DESIGN TITLE: LE4-7.pdsprj
DATE: 4/6/2026
8 Message(s)
ANIMATING: 00:00:57.600000 (CPU load 42%)





Part II

$$F_{min} = \frac{4000000}{4 \cdot 16 \cdot 256} = 244.14 \text{ Hz}$$

$$PR2 = \frac{F_{osc}}{4 \cdot F_{pwm} \cdot \text{Prescaler}} - 1$$

$$\text{10-bit Duty Value} = (\text{Duty Ratio}) \cdot 4 \cdot (PR2 + 1)$$

$$PR2 = \frac{4000000}{4 \cdot 500 \cdot 16} - 1 = 124$$

$$\text{Max Resolution} = 4 \cdot (124 + 1) = 500$$

$$10\%\ \text{Duty} = 500 \cdot 0.10 = 50$$

$$25\%\ \text{Duty} = 500 \cdot 0.25 = 125$$

$$50\%\ \text{Duty} = 500 \cdot 0.50 = 250$$

$$50\%\ \text{Duty} = 500 \cdot 0.50 = 250$$

$$95\%\ \text{Duty} = 500 \cdot 0.95 = 475$$

| ***Duty Cycle*** | ***10-bit Decimal Value*** | ***CCPR1L (8 MSB)*** | ***CCP1CON<5:4> (2 LSB)*** |
|---|---|---|---|
| *10%* | $10\%\ \text{Duty} = 500 \cdot 0.10 = 50$ | *12 (0x0C)* | *2* |
| *25%* | $25\%\ \text{Duty} = 500 \cdot 0.25 = 125$ | *31 (0x1F)* | *1* |
| *50%* | $50\%\ \text{Duty} = 500 \cdot 0.50 = 250$ | *62 (0x3E)* | *2* |
| *75%* | $50\%\ \text{Duty} = 500 \cdot 0.50 = 250$ | *93 (0x5D)* | *3* |

| *Duty Cycle* | *10-bit Decimal Value* | *CCPR1L (8 MSB)* | *CCP1CON<5:4> (2 LSB)* |
|---|---|---|---|
| *95%* | $95\%\ \text{Duty} = 500 \cdot 0.95 = 475$ | *118 (0x76)* | *3* |

$$PR2 = \frac{4000000}{4 \cdot 700 \cdot 16} - 1 = 88.28 \approx 88$$

$$\text{Max Resolution} = 4 \cdot (88 + 1) = 356$$

$$10\%\ \text{Duty} = 356 \cdot 0.10 = 35.6 \approx 35$$

$$25\%\ \text{Duty} = 356 \cdot 0.25 = 89$$

$$50\%\ \text{Duty} = 356 \cdot 0.50 = 178$$

$$75\%\ \text{Duty} = 356 \cdot 0.75 = 267$$

$$95\%\ \text{Duty} = 356 \cdot 0.95 = 338.2 \approx 338$$

| *Duty Cycle* | *10-bit Decimal Value* | *CCPR1L (8 MSB)* | *CCP1CON<5:4> (2 LSB)* |
|---|---|---|---|
| *10%* | $10\%\ \text{Duty} = 356 \cdot 0.10 = 35.6 \approx 35$ | *8 (0x08)* | *3* |
| *25%* | $25\%\ \text{Duty} = 356 \cdot 0.25 = 89$ | *22 (0x16)* | *1* |
| *50%* | $50\%\ \text{Duty} = 356 \cdot 0.50 = 178$ | *44 (0x2C)* | *2* |
| *75%* | $75\%\ \text{Duty} = 356 \cdot 0.75 = 267$ | *66 (0x42)* | *3* |
| *95%* | $95\%\ \text{Duty} = 356 \cdot 0.95 = 338.2 \approx 338$ | *84 (0x54)* | *2* |

$$PR2 = \frac{4000000}{4 \cdot 1000 \cdot 16} - 1 = 61.5 \approx 61$$

$$\text{Max Resolution} = 4 \cdot (61 + 1) = 248$$

$$10\%\ \text{Duty} = 248 \cdot 0.10 = 24.8 \approx 24$$

$$25\%\ \text{Duty} = 248 \cdot 0.25 = 62$$

$$50\%\ \text{Duty} = 248 \cdot 0.50 = 124$$

$$75\%\ \text{Duty} = 248 \cdot 0.75 = 186$$

$$95\%\ \text{Duty} = 248 \cdot 0.95 = 235.6 \approx 235$$

| *Duty Cycle* | *10-bit Decimal Value* | *CCPR1L (8 MSB)* | *CCP1CON<5:4> (2 LSB)* |
|---|---|---|---|
| *10%* | $10\%\ \text{Duty} = 248 \cdot 0.10 = 24.8 \approx 24$ | *6 (0x06)* | *0* |
| *25%* | $25\%\ \text{Duty} = 248 \cdot 0.25 = 62$ | *15 (0x0F)* | *2* |
| *50%* | $50\%\ \text{Duty} = 248 \cdot 0.50 = 124$ | *31 (0x1F)* | *0* |
| *75%* | $75\%\ \text{Duty} = 248 \cdot 0.75 = 186$ | *46 (0x2E)* | *2* |
| *95%* | $95\%\ \text{Duty} = 248 \cdot 0.95 = 235.6 \approx 235$ | *58 (0x3A)* | *3* |